UML Diagrams

Operating System File Explorer

Companion to the Software Requirements Specification (v1.0)

Use Case | Class | Sequence | Activity | State | Component

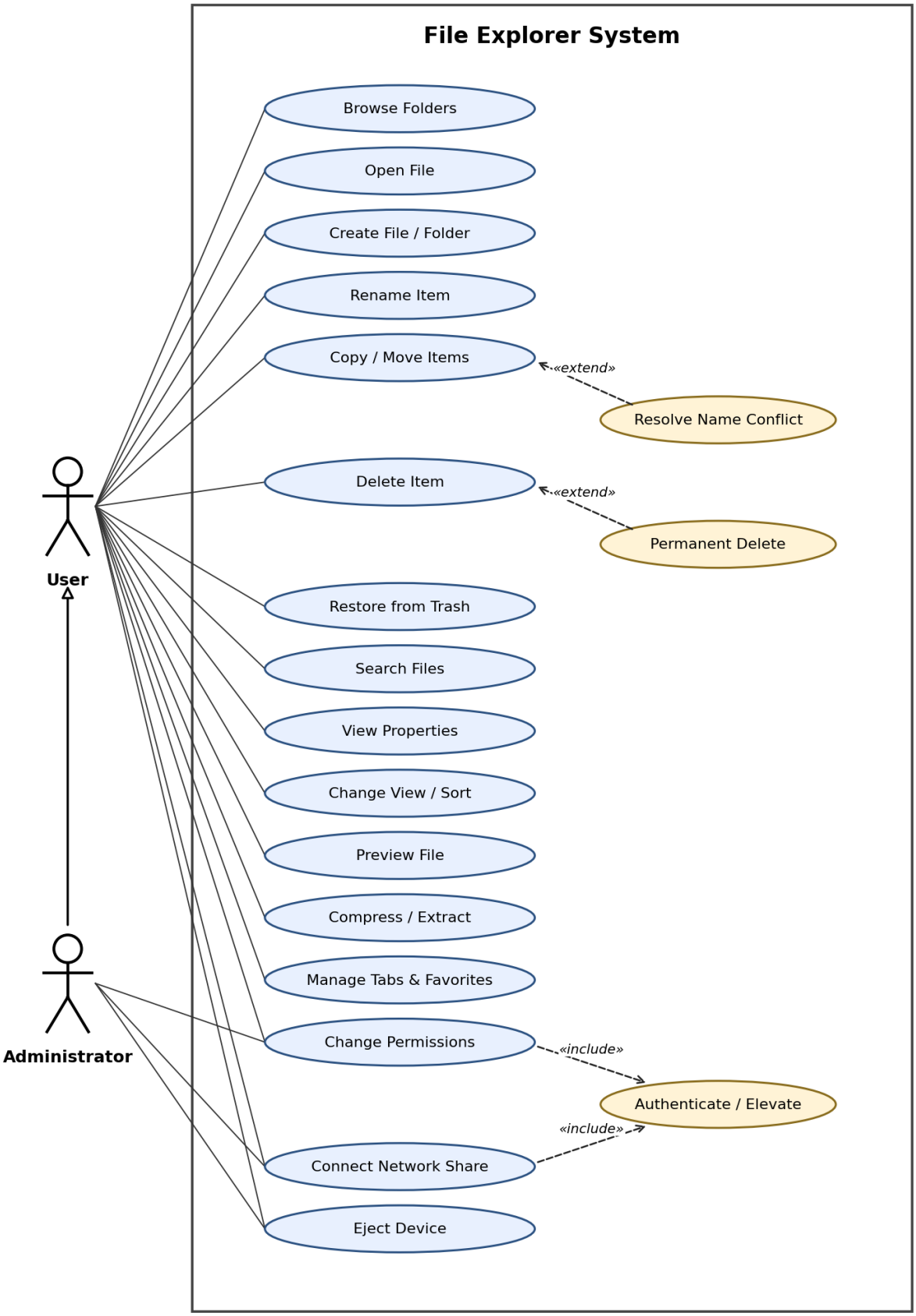
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Notation: solid arrow with hollow triangle = inheritance; dashed arrow with hollow triangle = realization; filled diamond = composition; hollow diamond = aggregation; dashed open arrow = dependency / use; *«extend»* and *«include»* = use case relationships.

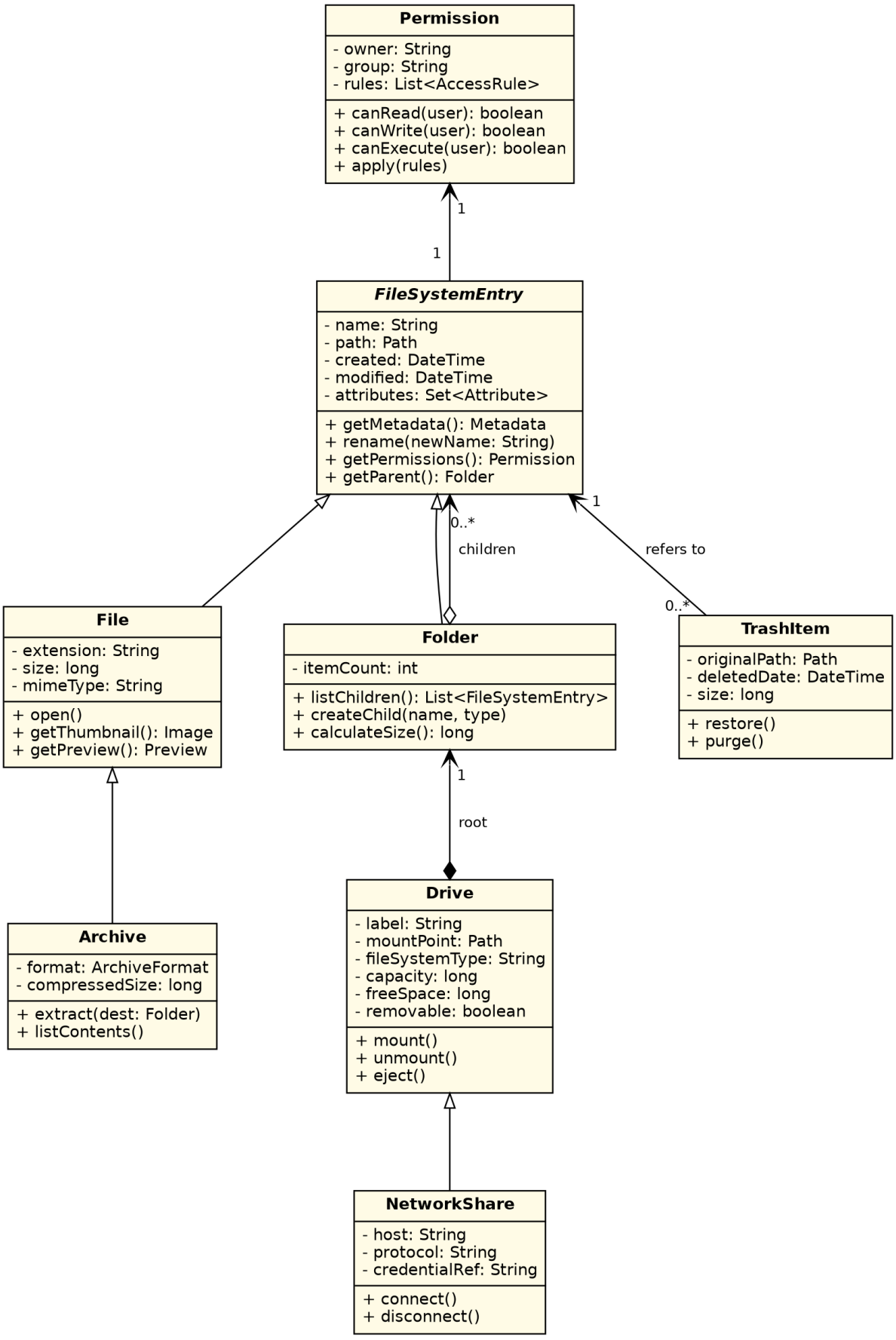
1. Use Case Diagram

Shows the main functions the File Explorer offers to its actors. The Administrator specializes the User and additionally manages permissions, network shares and device ejection. Dashed arrows show *extend* and *include* relationships.



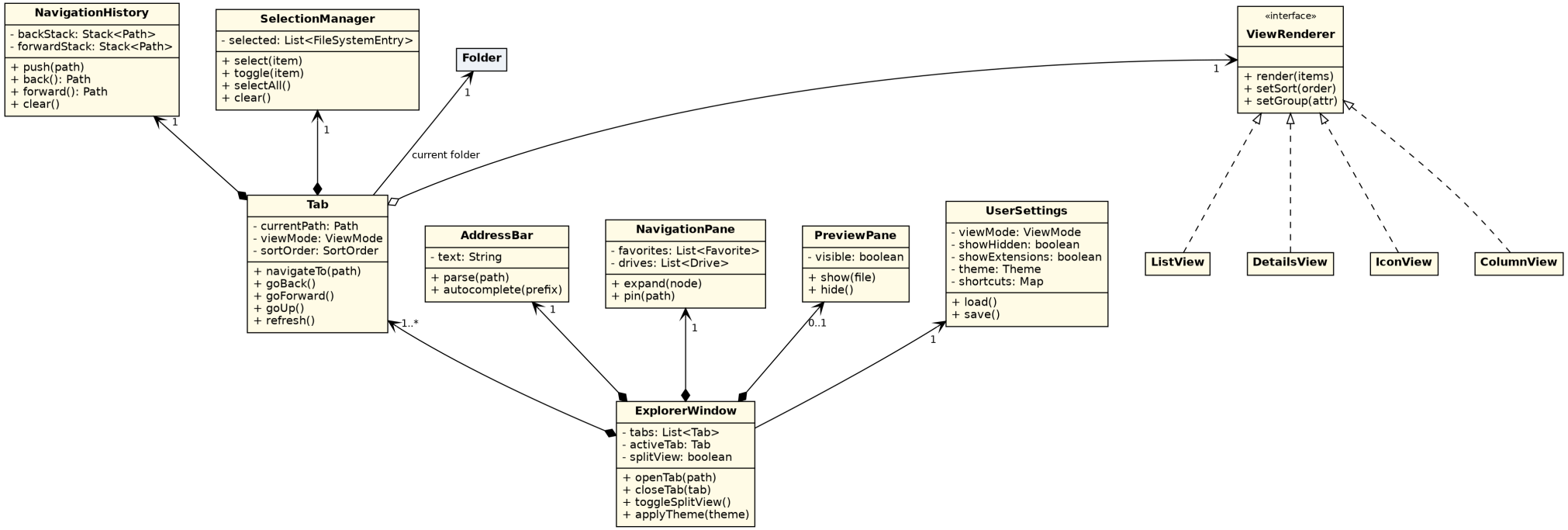
2. Class Diagram - Domain Model

Core data model. FileSystemEntry is the abstract base for File and Folder (a Folder aggregates its children). Each entry has a Permission object. Drive owns a root Folder; NetworkShare specializes Drive; TrashItem refers to a deleted entry.



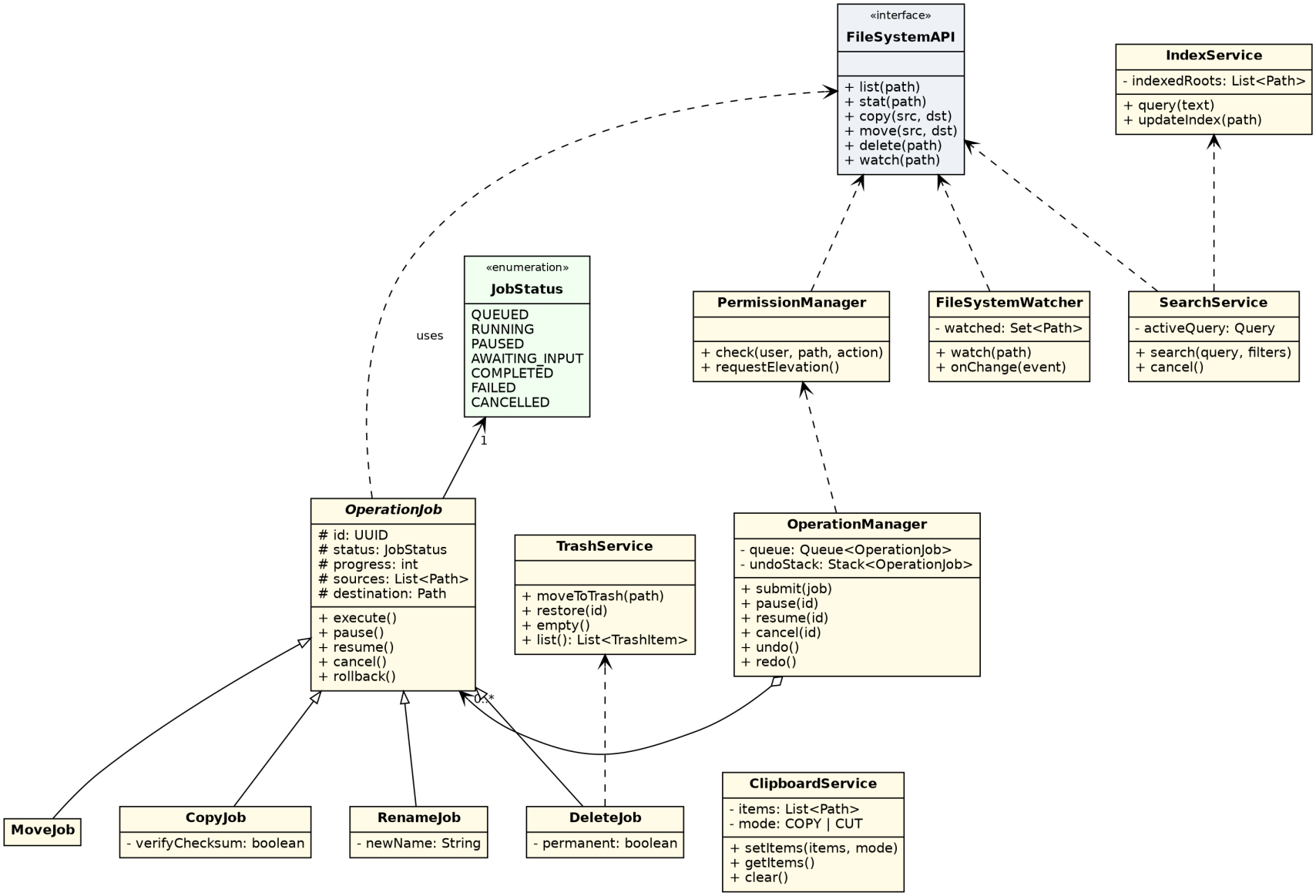
3. Class Diagram - UI and Navigation

Structure of the user interface. ExplorerWindow composes Tabs, AddressBar, NavigationPane and PreviewPane. Each Tab keeps its own navigation history, selection and a ViewRenderer (list, details, icon or column view).



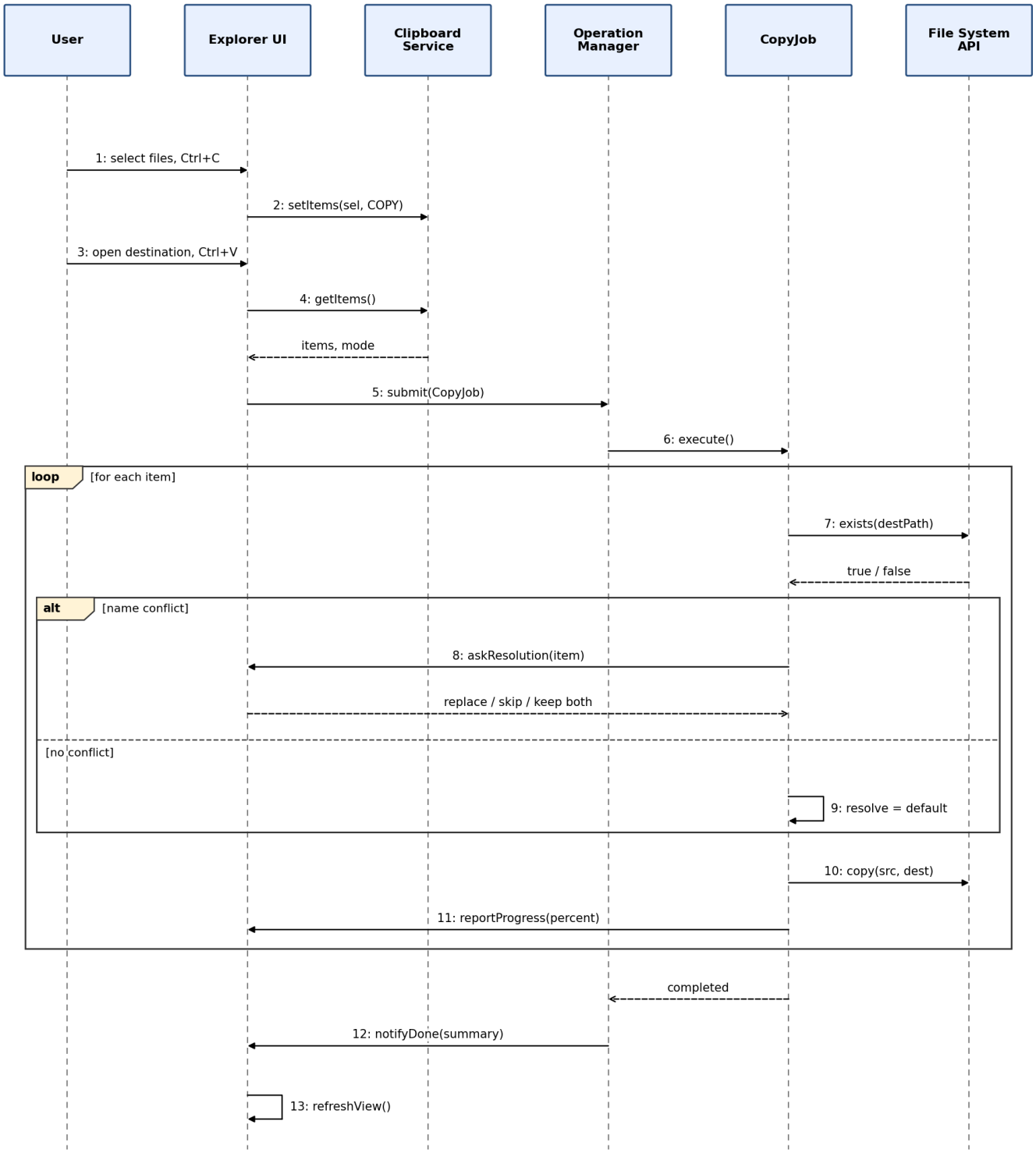
4. Class Diagram - Operations and Services

Background operations and supporting services. OperationManager queues OperationJobs (copy, move, delete, rename) that use the FileSystemAPI. Search, Trash, Permission and Watcher services complete the design.



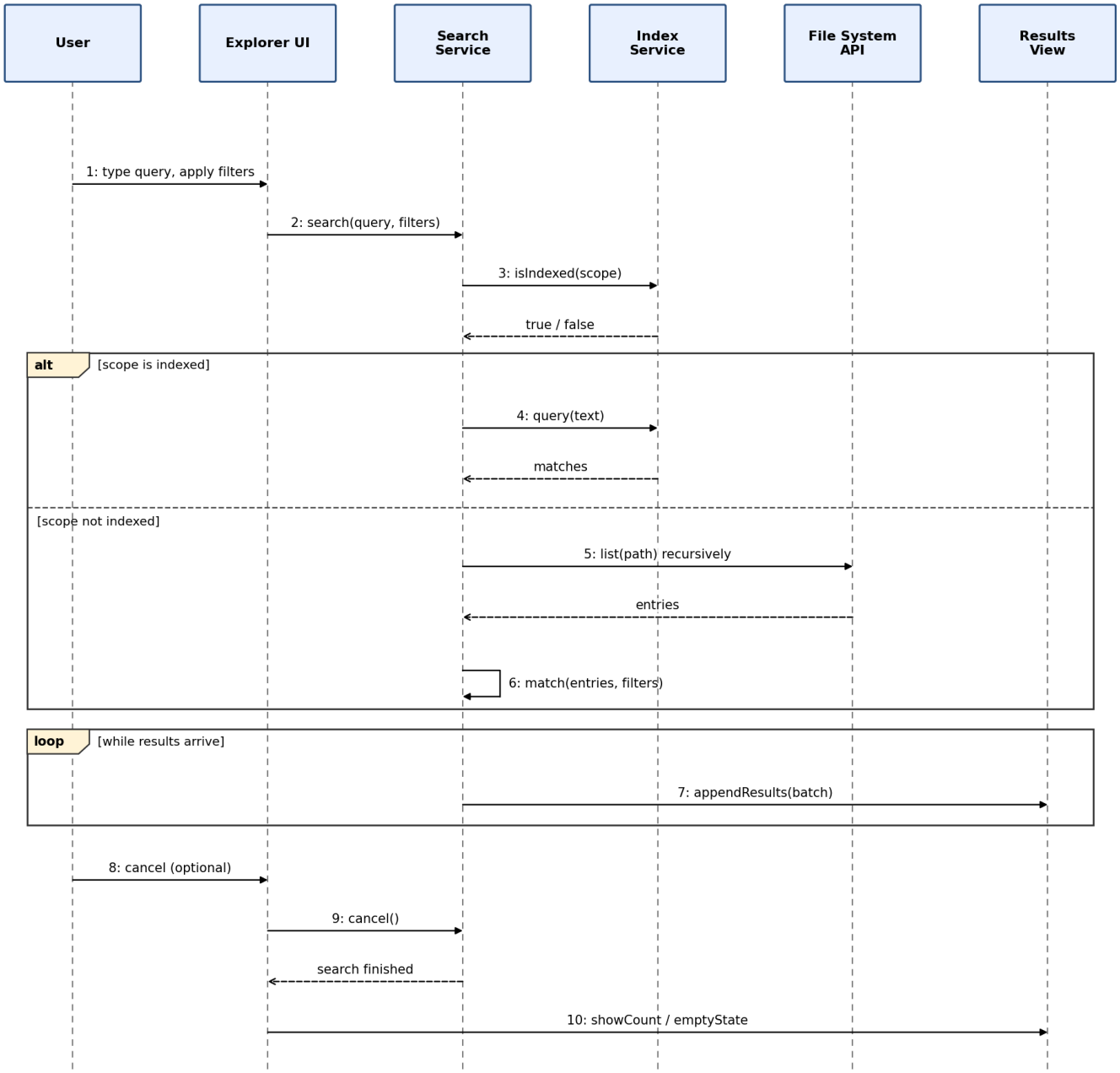
5. Sequence Diagram - Copy and Paste Files

Interaction between the user, UI, clipboard, operation manager and file system when files are copied. Includes the loop over items and the alternative path for name conflicts.



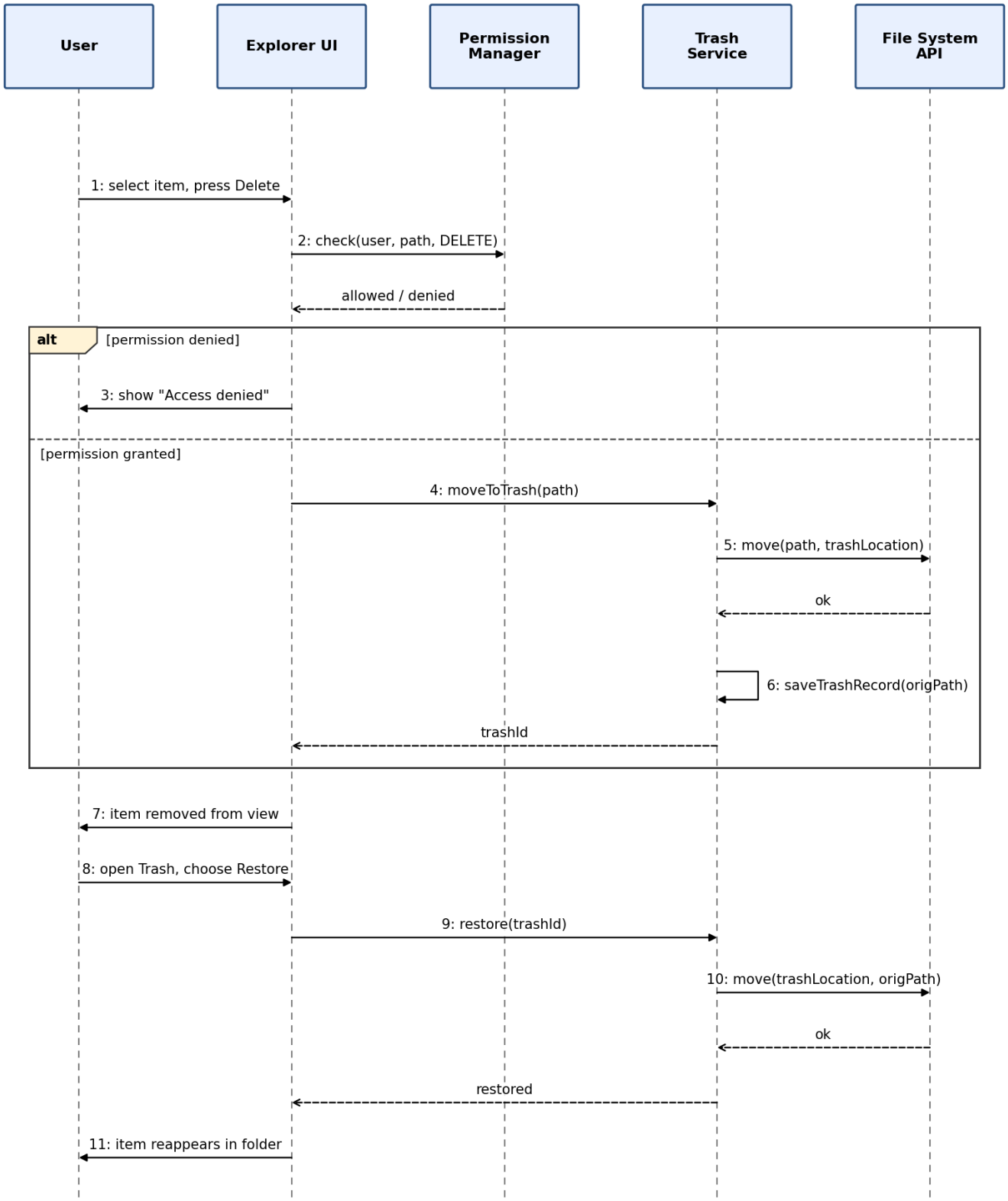
6. Sequence Diagram - Search Files

Search uses the index when the scope is indexed and falls back to scanning folders otherwise. Results are streamed to the view and the search can be cancelled at any time.



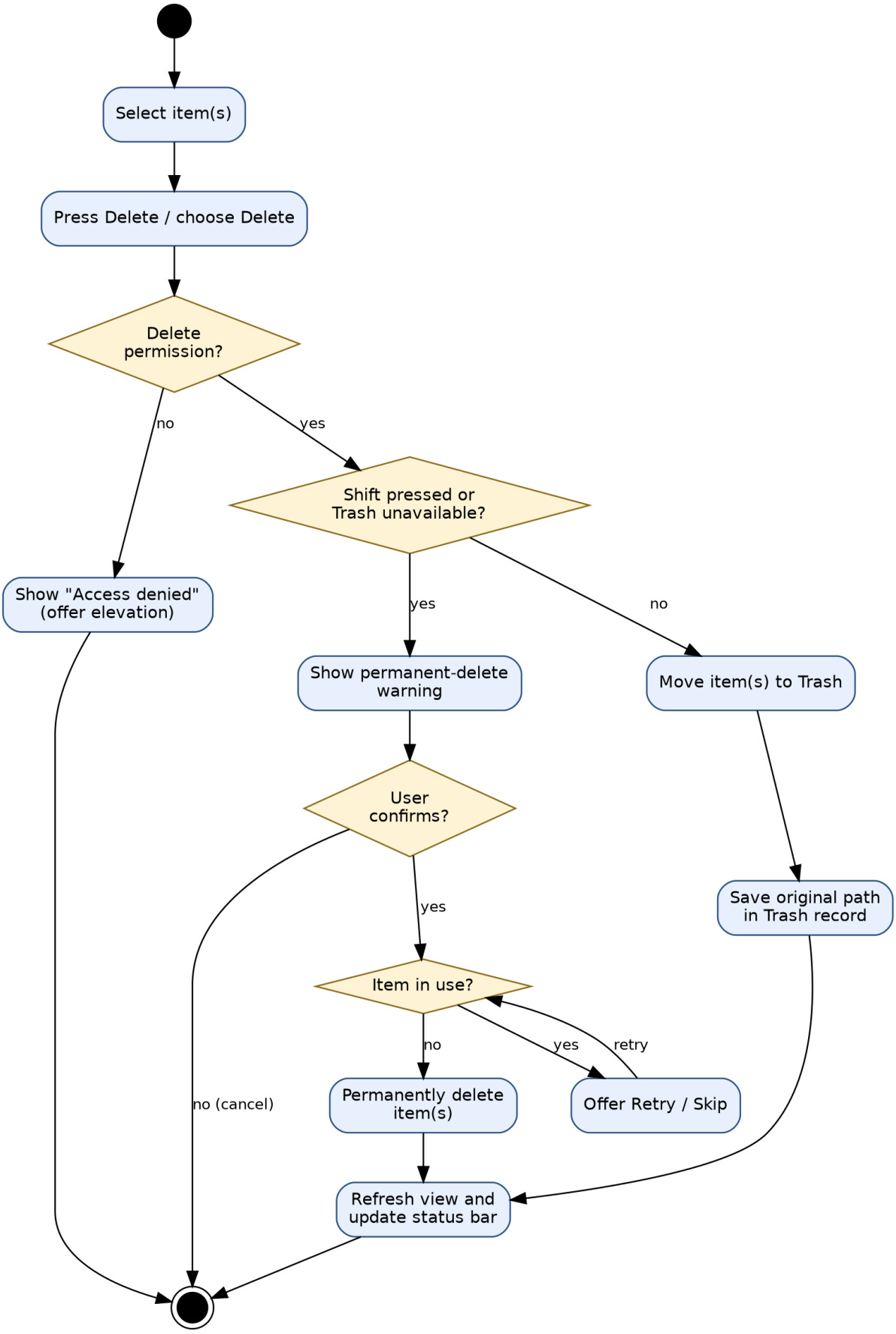
7. Sequence Diagram - Delete and Restore

Delete checks permissions, then moves the item to Trash and records its original path. Restore returns the item to its original location.



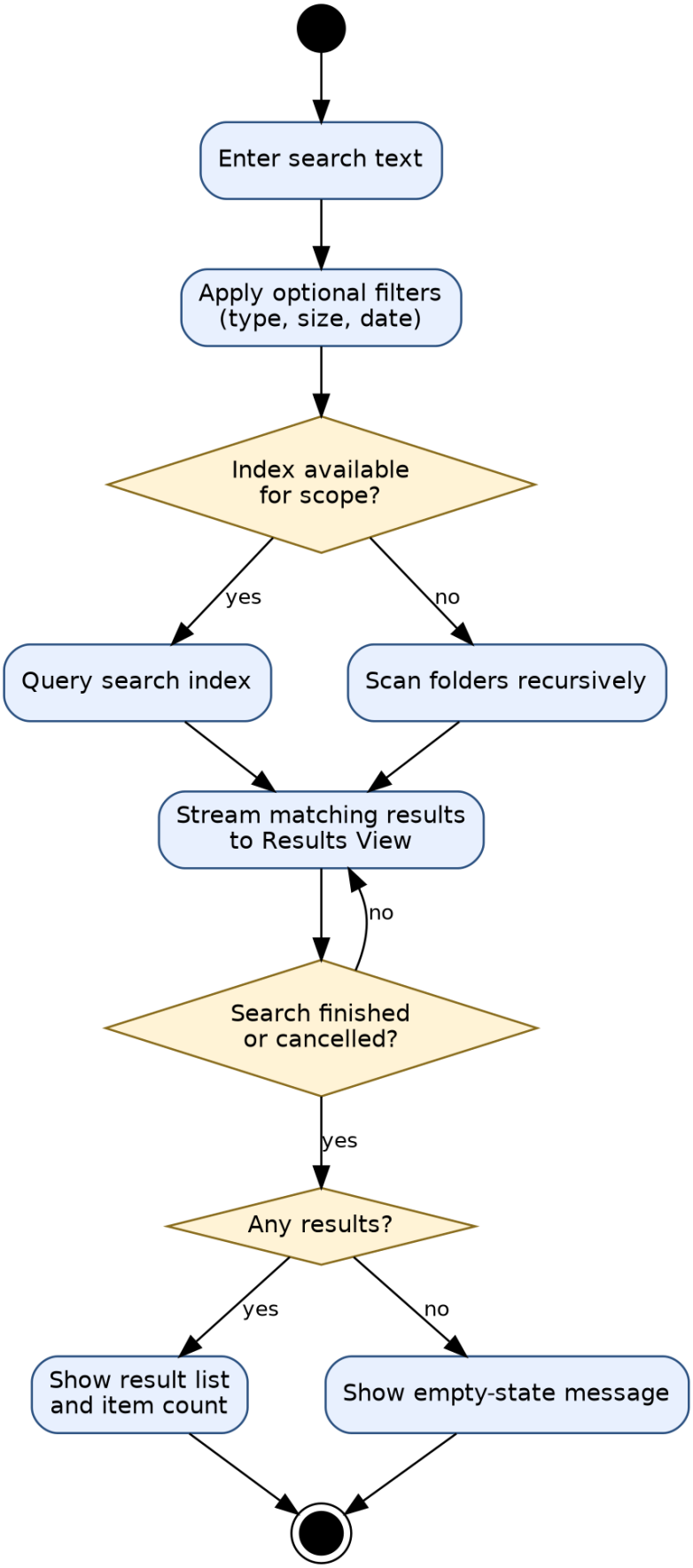
8. Activity Diagram - Delete Item

Workflow for deleting items, including permission check, permanent-delete confirmation and the item-in-use retry loop.



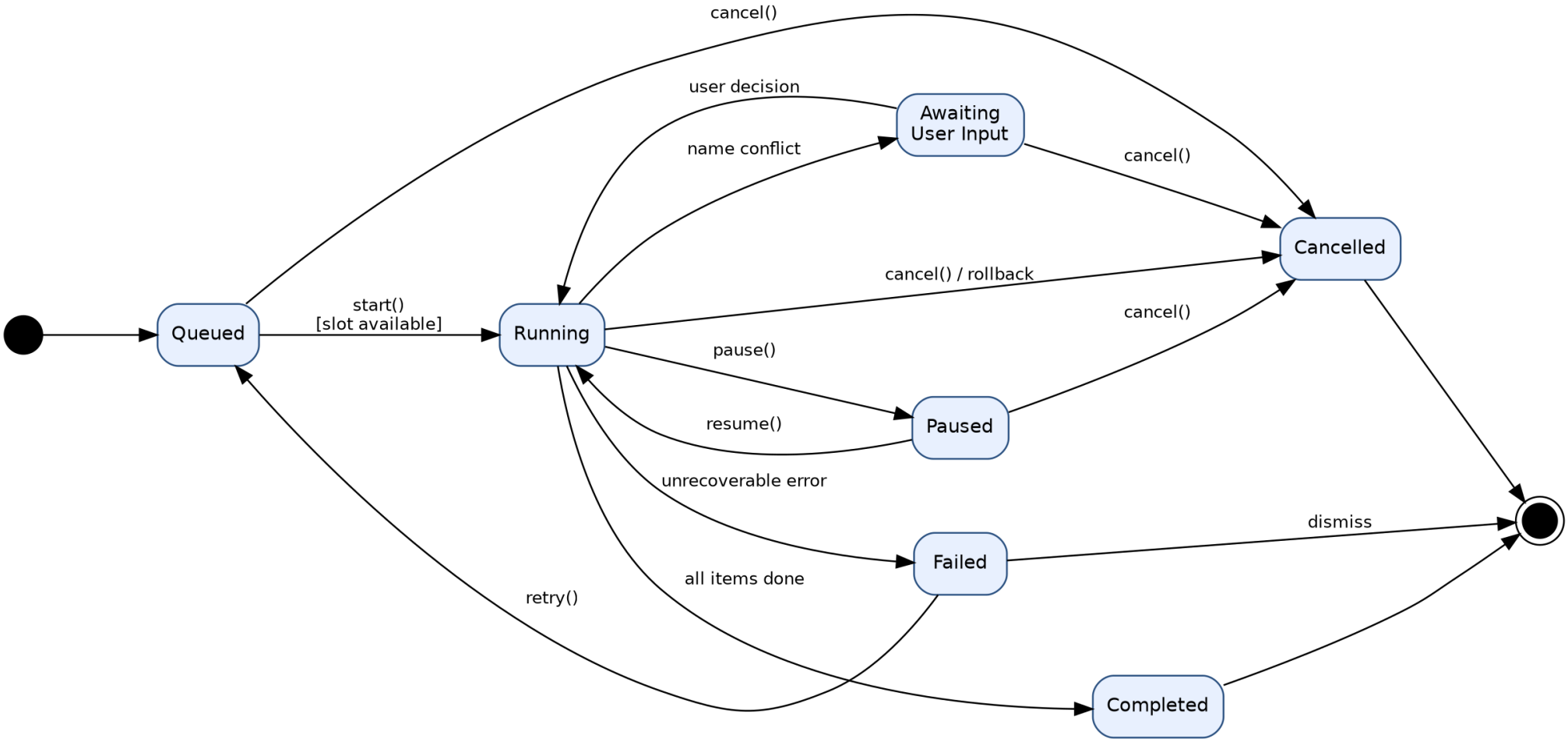
9. Activity Diagram - Search

Workflow of a search: choose index or recursive scan, stream results, then show results or an empty state.



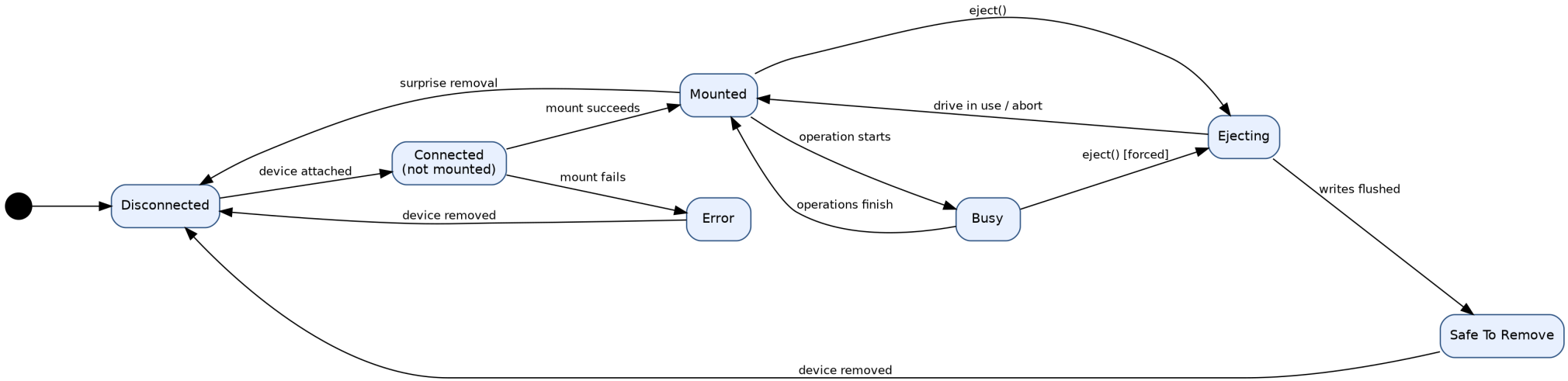
10. State Diagram - Operation Job

Lifecycle of a file operation job from queueing to completion, failure or cancellation, including pause/resume and waiting for user input on conflicts.



11. State Diagram - Drive

Lifecycle of a storage device from connection through mounting, use, ejection and safe removal.



12. Component Diagram

Layered architecture: presentation components depend on application controllers, which use services, which rely on platform APIs of the operating system.

